

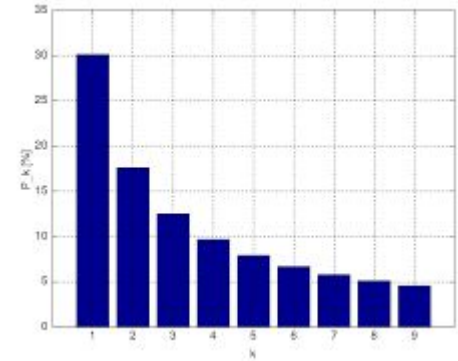


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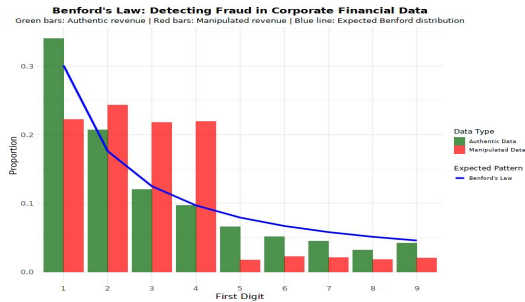
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Benford's Law

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We love making up numbers



Scenario

Mr. Dorst has been accused of embezzling school funds to buy a **gold toilet** for his private office. To hide the paper trail, he asks you (his "assistants") to forge 100 fake receipts for "office supplies."

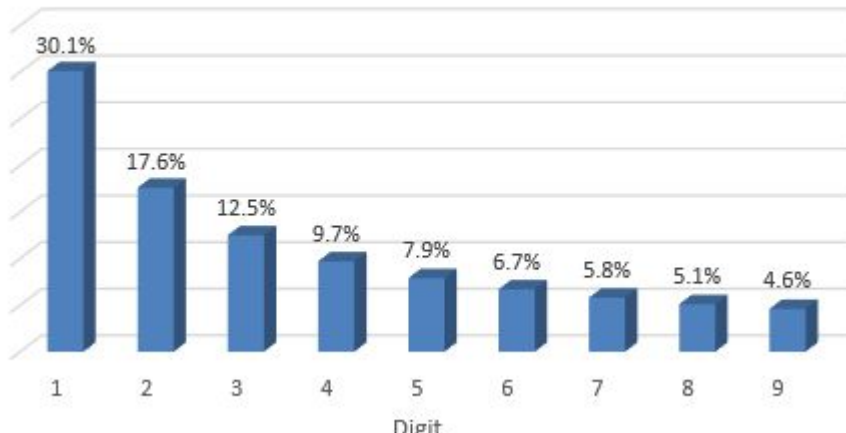


Benfords law

Benford's Law is the observation that in many real-life sets of numbers, smaller digits appear more often as the first digit. e.g. the digit **1** appears as the leading number much more frequently (about 30% of the time) than any other digit.

$$P(d) = \log_{10} \left(1 + \frac{1}{d} \right)$$

Benford's Law for Leading Digits

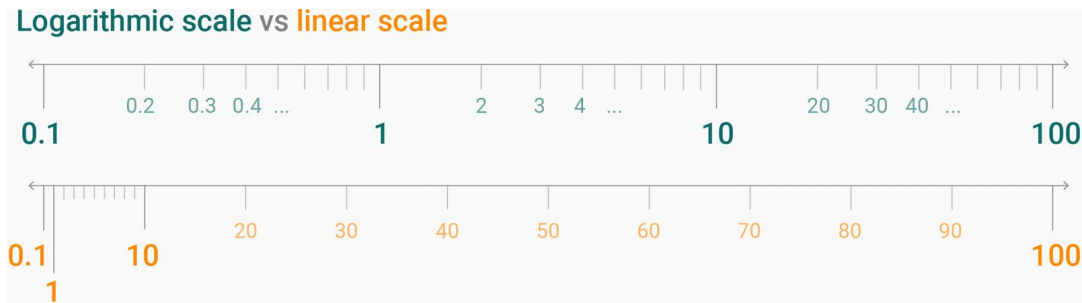


Why does work?

In nature, things rarely grow by adding a flat amount (like adding +1 every day). Instead, they grow by **percentages** (multiplication). This is the key to Benford's Law.

Imagine a small town with 1,000 people that grows at 10% per year:

- To get from **1,000 to 2,000** (a "1" leading digit), the town must add 1,000 people. At 10% growth, this takes about **7 years**.
- To get from **9,000 to 10,000** (a "9" leading digit), the town only needs to add 1,000 people. But because the town is already large, that 10% growth happens much faster—it takes **only 1 year**.



Requirements for benfords law

The data spans several "orders of magnitude" (e.g., numbers ranging from 10 to 1,000,000).

The data is not assigned: Zip codes, phone numbers, and Social Security numbers are "labels," not measurements, so they don't follow the law.

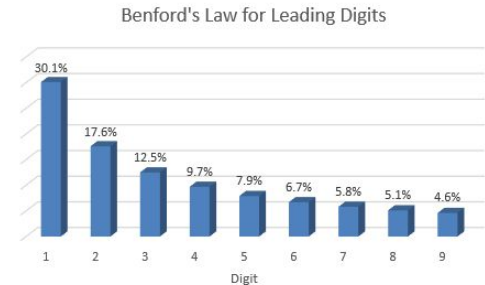
The data is not restricted: If you look at the heights of adult humans, almost everyone is between 1 and 2 meters tall. Therefore, the leading digit will almost *a/ways* be 1. This isn't Benford's Law; it's just biology.



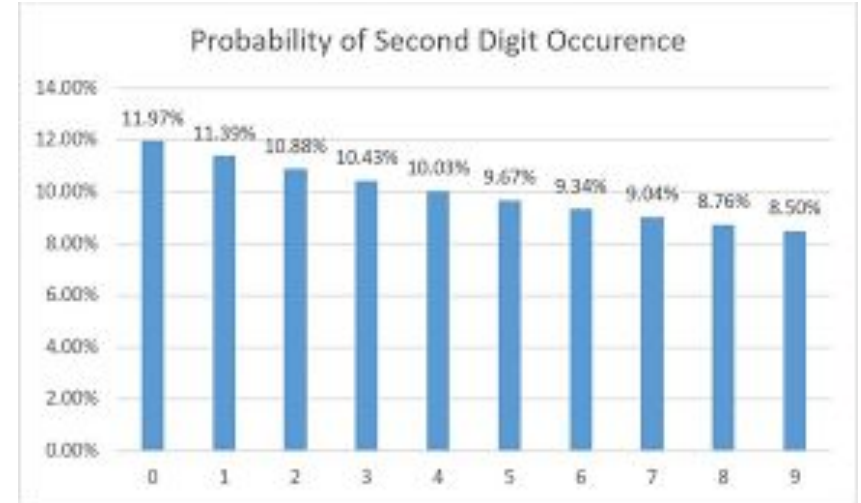
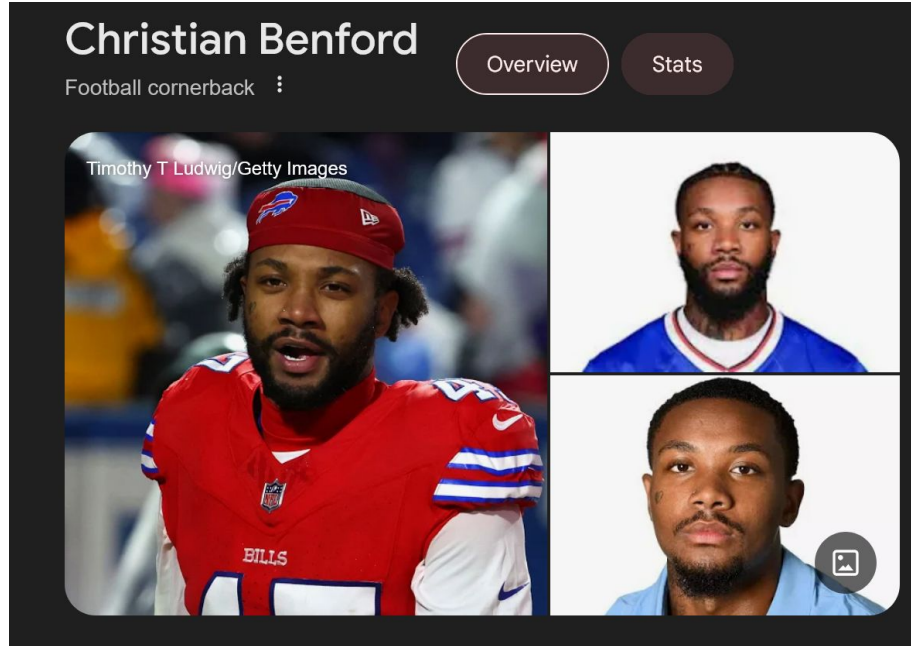
Scenario 2

Dorst is running for Supreme Dictator of the world" He knows about Benford's Law now, so he tells you to rig the vote counts from different voting machines. *"Make sure 30% of the vote counts start with a 1 so we don't get caught!"*

You create a list of votes (e.g., 122, 131, 105). You are careful to make the **first** digit follow Benford's Law.



Cooked by Benford again



Examples

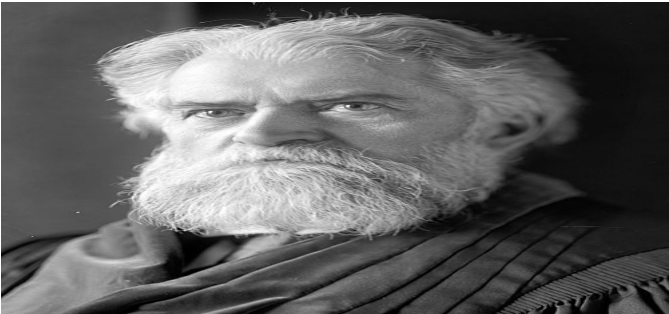


Leading digit	m		ft		Per Benford's law
	Count	Share	Count	Share	
1	23	39.7 %	15	25.9 %	30.1 %
2	12	20.7 %	8	13.8 %	17.6 %
3	6	10.3 %	5	8.6 %	12.5 %
4	5	8.6 %	7	12.1 %	9.7 %
5	2	3.4 %	9	15.5 %	7.9 %
6	5	8.6 %	4	6.9 %	6.7 %
7	1	1.7 %	3	5.2 %	5.8 %
8	4	6.9 %	6	10.3 %	5.1 %
9	0	0 %	1	1.7 %	4.6 %

$$\begin{aligned}2^1 &= 2 \\2^2 &= 4 \\2^3 &= 8 \\2^4 &= 16 \\2^5 &= 32 \\2^6 &= 64 \\2^7 &= 128 \\2^8 &= 256\end{aligned}$$

$$\begin{aligned}2^9 &= 512 \\2^{10} &= 1024 \\2^{11} &= 2048 \\2^{12} &= 4096 \\2^{13} &= 8192 \\2^{14} &= 16384 \\2^{15} &= 32768 \\2^{16} &= 65536\end{aligned}$$

Leading digit	Occurrence		Per Benford's law
	Count	Share	
1	29	30.2 %	30.1 %
2	17	17.7 %	17.6 %
3	12	12.5 %	12.5 %
4	10	10.4 %	9.7 %
5	7	7.3 %	7.9 %
6	6	6.3 %	6.7 %
7	5	5.2 %	5.8 %
8	5	5.2 %	5.1 %
9	5	5.2 %	4.6 %



Gr. 8

mm	Summa	1. Logarithmus	Differentia	Logarithmus	Summa
0	1000000	1000000	0	1000000	60
1	1000000	1000000	1	1000000	59
2	1000000	1000000	2	1000000	58
3	1000000	1000000	3	1000000	57
4	1000000	1000000	4	1000000	56
5	1000000	1000000	5	1000000	55
6	1000000	1000000	6	1000000	54
7	1000000	1000000	7	1000000	53
8	1000000	1000000	8	1000000	52
9	1000000	1000000	9	1000000	51
10	1000000	1000000	10	1000000	50
11	1000000	1000000	11	1000000	49
12	1000000	1000000	12	1000000	48
13	1000000	1000000	13	1000000	47
14	1000000	1000000	14	1000000	46
15	1000000	1000000	15	1000000	45
16	1000000	1000000	16	1000000	44
17	1000000	1000000	17	1000000	43
18	1000000	1000000	18	1000000	42
19	1000000	1000000	19	1000000	41
20	1000000	1000000	20	1000000	40
21	1000000	1000000	21	1000000	39
22	1000000	1000000	22	1000000	38
23	1000000	1000000	23	1000000	37
24	1000000	1000000	24	1000000	36
25	1000000	1000000	25	1000000	35
26	1000000	1000000	26	1000000	34
27	1000000	1000000	27	1000000	33
28	1000000	1000000	28	1000000	32
29	1000000	1000000	29	1000000	31
30	1000000	1000000	30	1000000	30

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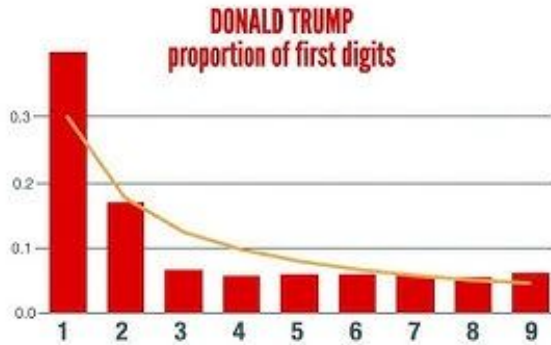
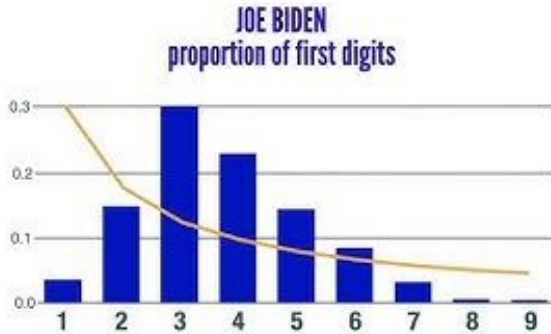
IRL benford aura

The Nelson Case: Forensic accountants used Benford's Law to catch Wayne James Nelson, an Arizona state employee who embezzled nearly **\$2 million** by issuing 23 checks with bogus, human-made amounts that didn't follow natural digit frequencies.

The Enron Scandal: After the energy giant collapsed, researchers analyzed Enron's financial statements and found massive deviations from Benford's Law, proving the company had been "**cooking the books**" with fake numbers for years.

Greek Economic Crisis: The European Union used Benford's Law to analyze Greek macroeconomic data and discovered that the country's reported deficit figures had likely been **manipulated** to meet EU targets.

Joe Biden rigged the election?



Joe Biden rigged the election

no

